

Absolute and conditional convergence - 5.5

Definition: We say that a series $\sum_{n=1}^{\infty} a_n$ converges absolutely if $\sum_{n=1}^{\infty} |a_n|$ converges. We say it converges conditionally if it converges but $\sum_{n=1}^{\infty} |a_n|$ doesn't.

Example: $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$ converges by the AST, but

$\sum_{n=1}^{\infty} |(-1)^{n-1} \frac{1}{n}| = \sum_{n=1}^{\infty} \frac{1}{n}$ diverges (p-test, $p=1$). Thus

$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$ converges conditionally.

Example: $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{3n+1}{n^3+9}$ converges absolutely,

because $|(-1)^{n-1} \frac{3n+1}{n^3+9}| = \frac{3n+1}{n^3+9}$, and

$$\lim_{n \rightarrow \infty} \frac{\frac{3n+1}{n^3+9}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{3n^2+n^2}{n^3+9} = 3 \quad \text{which is nonzero,}$$

so by the LCT, the fact that $\sum_{n=1}^{\infty} \frac{1}{n^2}$

converges (p-test, $p=2$) implies that $\sum_{n=1}^{\infty} \frac{3n+1}{n^3+9}$

converges as well.

Proposition: If $\sum_{n=1}^{\infty} |a_n|$ converges, then
so does $\sum_{n=1}^{\infty} a_n$. (Absolute convergence implies convergence.)

pf. Observation: $a_n = (a_n + |a_n|) - |a_n|$.

Note also that $0 \leq a_n + |a_n| \leq 2|a_n|$.

Then $\sum_{n=1}^{\infty} 2|a_n|$ converges, so by the DCT,

so does $\sum_{n=1}^{\infty} a_n + |a_n|$. Thus

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_n + |a_n| - \sum_{n=1}^{\infty} |a_n| \text{ converges. } \square$$

"Analyze the convergence" of a series.

→ Say whether it diverges, converges
absolutely, or converges conditionally.

requires showing two things:

- original series converges
- $|a_n|$ series diverges

Example: Analyze the convergence of $\sum_{n=1}^{\infty} \frac{\cos(n)}{n^2}$.

- Divergence test doesn't tell us anything
since $\lim_{n \rightarrow \infty} \frac{\cos(n)}{n^2} = 0$.

- Consider $\left| \frac{\cos(n)}{n^2} \right| \leq \frac{1}{n^2}$. We know by
the p-test ($p=2$) that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

so $\sum_{n=1}^{\infty} \left| \frac{\cos(n)}{n^2} \right|$ also converges by the DCT.

- Thus $\sum_{n=1}^{\infty} \frac{\cos(n)}{n^2}$ converges absolutely.

Example: Analyze the convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$.

- $\left| \frac{(-1)^n n}{n^2+1} \right| = \frac{n}{n^2+1} \rightarrow 0$ as $n \rightarrow \infty$, so this passes the divergence test.

- $\frac{n}{n^2+1}$ decreases to 0, and $\frac{(-1)^n n}{n^2+1}$ has terms with alternating signs, so by the AST, $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$ converges.

- $\lim_{n \rightarrow \infty} \frac{\frac{n}{n^2+1}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = 1$, so by the LCT,

$\sum_{n=1}^{\infty} \frac{n}{n^2+1}$ diverges since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (p-test, $p=1$).

- Thus $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$ converges conditionally.